

(1001CJA102120051)

Test Pattern



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Main)
FULL SYLLABUS
08-02-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	C	A	D	A	C	B	B	D
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	A	D	B	A	A	D	D	D	B
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1.00	3.75	1.05	0.45	1.10	4.00	0.30	3.10 to 3.12	2.25	3.73

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	C	D	C	B	A	C	B	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	A	D	B	D	A	B	B	A	B
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5.00	1.28	3.00	25.50	9.22 to 9.24	14.00	8.00	9.20	3.00	8.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	A	C	B	C	A	C	B	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	D	D	A	D	D	B	A	C	C
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	7.00	833.33	10.00	5.00	3.00	200.00	6.00	34.00	5.00	9.00



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PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (B)

Sol. $U_i = \frac{2kQq}{a} + \frac{kq^2}{2a}$

$$U_f = kQq \left(\frac{1}{a+x} + \frac{1}{a-x} \right) + \frac{kq^2}{2a}$$

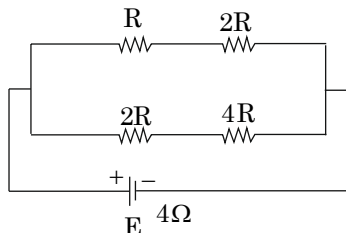
Hence, $k = \frac{1}{4\pi\epsilon_0}$

$$|\Delta U| = |U_f - U_i| = \frac{2kQqx^2}{a^3} \text{ (for } x \ll a \text{)}$$

$$\therefore \Delta U \propto x^2$$

2. Ans. (B)

Sol. Given circuit is a balanced wheat stone bridge, so no current flows through 6R resistor and equivalent circuit is



So, external resistance

$$= R_{eq} = \frac{6R \times 3R}{6R + 3R} = 2R$$

For maximum power output, when

$$R_{eq} = 4\Omega$$

$$\Rightarrow 2R = 4 \Rightarrow R = 2\Omega$$

3. Ans. (C)

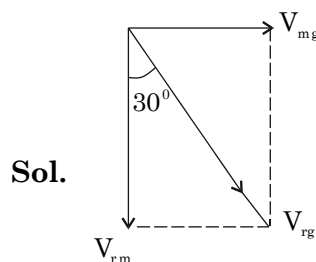
Sol. $\frac{1}{2}m_1u_1^2 - \frac{1}{2}m_1v_1^2 = \frac{75}{100} \times \frac{1}{2}m_1u_1^2$

$$\Rightarrow u_1^2 - v_1^2 = \frac{3}{4}u_1^2 \Rightarrow v_1 = \frac{1}{2}u_1$$

$$\text{Also, } v_1 = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) u_1 \Rightarrow \frac{1}{2}u_1 = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) u_1$$

$$\Rightarrow m_2 = 3m_1 = 3m$$

4. Ans. (A)



Sol.

$$V_{rg} = \frac{V_{mg}}{\sin 30^\circ} = \frac{10}{1/2}$$

$$= 20 \text{ km/h}$$

5. Ans. (D)

Sol. $C_p = \frac{Q}{n\Delta\theta} = \frac{70}{2 \times 5} = 7 \text{ cal mol}^{-1} \text{K}^{-1}$

$$C_v = C_p - R = 7 - 2 = 5 \text{ cal mol}^{-1} \text{K}^{-1}$$

Amount of heat required at constant volume

$$Q = nC_v\Delta\theta = 2 \times 5 \times 5 = 50 \text{ cal}$$

6. Ans. (A)

Sol. $\beta = \frac{\lambda D}{d} = 0.25 \times 10^{-3} \Rightarrow \lambda D = \frac{d}{4} \times 10^{-3} \dots (1)$

$$\beta' = \frac{\lambda D}{d + 1.2 \times 10^{-3}} = \frac{\beta}{1.5} = \frac{0.25 \times 10^{-3}}{1.5} = \frac{10^{-3}}{6}$$

$$\Rightarrow \lambda D = \frac{10^{-3}}{6} (d + 1.2 \times 10^{-3}) \dots (2)$$

From eq = (1) and (2), $\frac{d}{4} = \frac{d}{6} + \frac{1.2 \times 10^{-3}}{6}$

$$\Rightarrow d \left(\frac{1}{4} - \frac{1}{6} \right) = \frac{1.2 \times 10^{-3}}{6}$$

$$\Rightarrow d = 2.4 \times 10^{-3} \text{ m}$$

From (1) : $\lambda = \frac{d \times 10^{-3}}{4D} = \frac{2.4 \times 10^{-3} \times 10^{-3}}{4 \times 1}$
 $= 0.6 \mu\text{m}$

7. Ans. (C)

Sol. $T = 2\pi\sqrt{\frac{l}{g}}$

If effective acceleration due to gravity

$$= g - \frac{3g}{4} = \frac{g}{4}$$

New time period $= T' = 2\pi\sqrt{\frac{l}{g/4}}$

$$= 2\left[2\pi\sqrt{\frac{l}{g}}\right] = 2T$$

8. Ans. (B)

Sol. Velocity at the lowest point,

$$v = \sqrt{\frac{2gh}{1 + \frac{K^2}{R^2}}}$$

For solid sphere, $\frac{K^2}{R^2} = \frac{2}{5}$

$$v = \sqrt{\frac{2 \times 10 \times 7}{1 + \frac{2}{5}}} = 10 \text{ m/s}$$

9. Ans. (B)

Sol. $E \propto AT^4$

$$E \propto r^2$$

$$\therefore \frac{E_1}{E_2} = \frac{r_1^2}{r_2^2} = \left(\frac{2}{1}\right)^2 = \frac{4}{1}$$

10. Ans. (D)

Sol. $\beta_1 = 10 \log_{10} \left(\frac{I}{I_0} \right)$ and $\beta_2 = 10 \log_{10} \left(\frac{I'}{I_0} \right)$

given, $\beta_1 - \beta_2 = 20$

$$\therefore 20 = 10 \log \left(\frac{I}{I'} \right)$$

$$\Rightarrow I' = \frac{I}{100}$$

11. Ans. (A)

Sol. $\frac{n_B}{n_A} = \frac{P_B V_B / RT}{P_A V_A / RT} = \frac{P_B V_B}{P_A V_A}$

$$= \frac{\left(P + \frac{4s}{r_B} \right) \times \frac{4}{3} \pi r_B^3}{\left(P + \frac{4s}{r_A} \right) \times \frac{4}{3} \pi r_A^3}$$

Substituting the values, $\frac{n_B}{n_A} = 6$

12. Ans. (A)

Sol. Power of combination, $p = p_1 + p_2$

$$= 3 - 5 = -2D$$

$$p = \frac{100}{f} \Rightarrow f = \frac{100}{p} = -50 \text{ cm}$$

From lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{-50} = \frac{1}{v} - \frac{1}{-50}$$

$$\Rightarrow v = -25 \text{ cm}$$

13. Ans. (D)

Sol. Max force applied on B, till both

$$\text{move together } F_{\max} = \mu m_A g \left(1 + \frac{m_B}{m_A} \right)$$

$$= 0.5 \times 2 \times 10 \left(1 + \frac{8}{2} \right)$$

$$= 10(1 + 4) = 50 \text{ N}$$

applied force, $F = 20 \text{ N} < F_{\max}$

$$\therefore \text{both moves together, } a = \frac{F}{m_A + m_B}$$

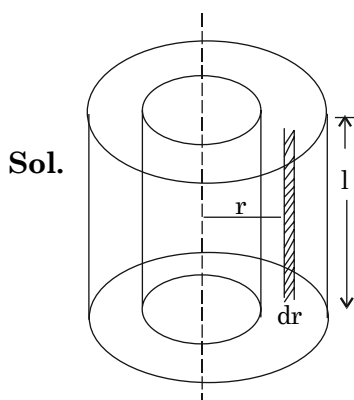
$$= 20 / (2 + 8)$$

$$= 2 \text{ m/s}^2$$

$$\therefore \text{friction force on A} = f_A = m_A a$$

$$= 2 \times 2 = 4 \text{ N}$$

14. Ans. (B)



$$R_{eq} = \int_{r_1}^{r_2} \frac{dr}{K \cdot 2\pi r l} = \frac{\ln 2}{K \cdot 2\pi l}$$

$$\text{Heat current} = \frac{T_s - T_{ice}}{R_{eq}}$$

$$= \frac{40}{\ln 2} K(2\pi l) = 80\pi \text{ J/sec}$$

$$\text{Also, } \frac{dm_{ice}}{dt} L_f = 80\pi$$

$$\Rightarrow \frac{dm_{ice}}{dt} = \frac{80\pi}{80 \times 4.2 \times 1000} = \frac{\pi}{4200} \text{ kg/sec}$$

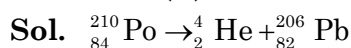
15. Ans. (A)

$$\text{Sol. Radiation pressure} = p = \frac{2 \times \left(\frac{1000}{0.5} \right) \left(\frac{1}{2} \right)}{3 \times 10^8}$$

$$= 0.67 \times 10^{-5} \text{ N/m}^2$$

As only $\frac{1}{2}$ power is pushing to the astronaut.

16. Ans. (A)



$$\text{mass defect} = \Delta m = (m_{po} - m_{He} - m_{Pb})$$

$$= 0.005818u$$

$$\therefore Q = (\Delta m) \times 931.48 \text{ MeV} = 5419 \text{ keV}$$

From conservation of linear momentum,

$$P_b = P_\alpha$$

$$\therefore \sqrt{2m_{Pb}K_{Pb}} = \sqrt{2m_\alpha K_\alpha} \Rightarrow \frac{K_\alpha}{K_{Pb}} = \frac{m_{Pb}}{m_\alpha} = \frac{206}{4}$$

$$\therefore K_\alpha = (Q) \left(\frac{206}{206+4} \right) \quad (K_{total} = Q)$$

$$= \frac{206}{210} \times 5419 \approx 5319 \text{ keV}$$

17. Ans. (D)

Sol. If temperature $T < T_c$ magnetic field inside conduction = 0.

18. Ans. (D)

Sol. Given, $R = 3\Omega$, $X_L = 4\Omega$

$$\text{In L-R circuit, } \tan \phi = \frac{X_L}{R} = \frac{4}{3} \Rightarrow \phi = 53^\circ$$

Current across inductor

$$I = I_0 \sin(\omega t - 53^\circ) = \frac{V_0}{Z} \sin(\omega t - 53^\circ)$$

$$= \frac{10}{\sqrt{3^2 + 4^2}} \sin(\omega t - 53^\circ) = 2 \sin(\omega t - 53^\circ)$$

$\therefore$ voltage across the inductor at $t = T/2$

$$(V_L)_0 = I_0 X_L = 2 \times 4 = 8$$

$$V_L = (V_L)_0 \sin(\omega T/2 - 53^\circ + \pi/2)$$

$$= 8 \sin(\pi - 53^\circ + \pi/2)$$

$$|V_L| = 8 \sin 37^\circ = 8 \times \frac{3}{5} = 4.8 \text{ V}$$

19. Ans. (D)

$$\text{Sol. } a \leq r \leq b \quad E(r) = \frac{q}{4\pi\epsilon_0 kr^2}$$

$$b \leq r \leq c \quad E(r) = \frac{q}{4\pi\epsilon_0 r^2}$$

So, potential difference between A and C

$$\Delta V = V_A - V_B$$

$$= - \int_a^b \frac{q}{4\pi\epsilon_0 kr^2} dr - \int_b^c \frac{q}{4\pi\epsilon_0 r^2} dr$$

$$= \frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{k} \left(\frac{1}{a} - \frac{1}{b} \right) + \left(\frac{1}{b} - \frac{1}{c} \right) \right\}$$

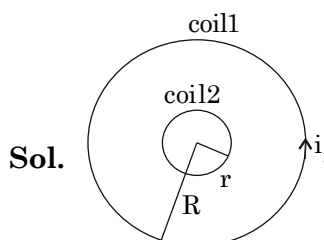
$$= \frac{q}{4\pi\epsilon_0} \left(\frac{b-a}{kab} + \frac{c-a}{bc} \right)$$

$$= \frac{q}{4\pi\epsilon_0 kabc} (c(b-a) + ka(c-b))$$

capacitance of system is

$$C = \frac{q}{\Delta V} = \frac{4\pi\epsilon_0 kabc}{ka(c-b) + c(b-a)}$$

20. Ans. (B)



Sol.

$$\text{Flux across coil 2} = \phi_2 = M i_1 \dots (1)$$

$$\phi_2 = B_1 A_2 \dots (2)$$

$$\frac{\mu_0 i_1}{2R} \pi r^2 = M i_1 \Rightarrow M = \frac{\mu_0 \pi r^2}{2R}$$

SECTION-II

1. **Ans. 1.00**

Sol. Centre of mass of binary star lies at centre of their circular path.

$$M_1 r_1 = M_2 r_2$$

$$\text{Then, } G \frac{M_1 M_2}{(r_1 + r_2)^2} = M_1 \omega^2 r_1 = M_2 \omega^2 r_2$$

$$\Rightarrow \omega_1 = \omega_2 \Rightarrow \frac{T_1}{T_2} = 1.00$$

Same angular velocity and time period of revolution.

2. **Ans. 3.75**

Sol. From equation of motion $v^2 = u^2 + 2as$

$$\Rightarrow (5000)^2 = (1000)^2 - 2as \Rightarrow s = \frac{37500}{a}$$

workdone against air resistance

$$= ma \times s = \frac{10}{1000} \times 375000$$

$$= 3750 \text{ Joule} = 3.75 \text{ kJ}$$

3. **Ans. 1.05**

$$\text{Sol. } 315 = \frac{nv}{2L} \text{ and } 420 = \frac{(n+1)v}{2L}$$

$$\text{divide, } \frac{315}{420} = \frac{n}{n+1} \Rightarrow n = 3$$

$$\text{lowest frequency} = \frac{v}{2L} = \frac{315}{3} = 105 \text{ Hz}$$

$$\text{Then, } x = 1.05$$

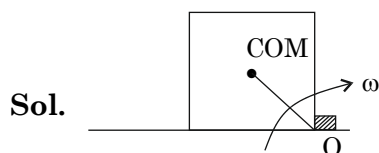
4. **Ans. 0.45**

Sol. induced emf

$$= e = B_H l v = 5 \times 10^{-5} \times 3 \times 3$$

$$= 0.45 \times 10^{-3} = 0.45 \text{ mV}$$

5. **Ans. 1.10**



$$M v_0 \left(\frac{a}{2} \right) = I_0 \omega \text{ (apply COAM)}$$

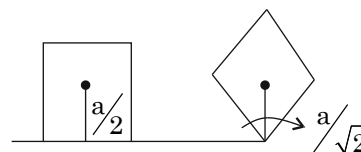
$$= \left(\frac{Ma^2}{6} + M \left(\frac{a}{\sqrt{2}} \right)^2 \right) \omega \Rightarrow \omega = \frac{3v_0}{4a}$$

Now, from conservation of energy.

$$\frac{1}{2} I \omega^2 = mg(\Delta h)$$

$$\Rightarrow \frac{1}{2} \left(\frac{2}{3} Ma^2 \right) \left(\frac{3v_0}{4a} \right)^2 = Mg \left(\frac{a}{\sqrt{2}} - \frac{a}{2} \right)$$

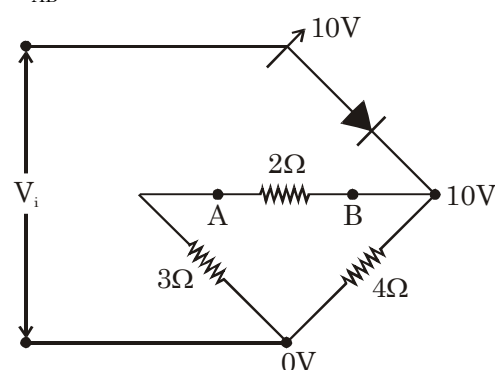
$$\Rightarrow v_0 = \sqrt{1.1} ag$$



6. **Ans. 4.00**

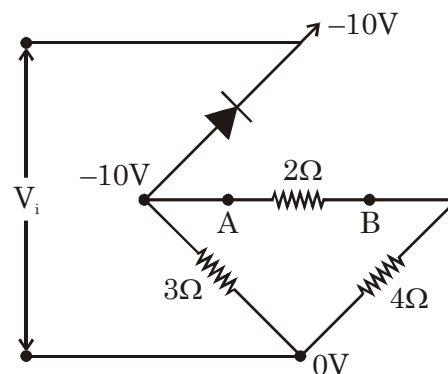
Sol. For upper half

$$V_{AB} = 4 \text{ V}$$



For lower half

$$V_{AB} = \frac{10}{6} \times 2 = 3.33 \text{ V}$$



So maximum voltage = 4V

7. **Ans. 0.30**

Sol. Initially only Ge will work

$$i = \frac{15.2}{8} \quad (V_0)_i = \frac{15.2}{8} \times 6$$

Finally only Si will work

$$i = \frac{14.8}{8} \quad (V_0)_f = \frac{14.8}{8} \times 6$$

$$\text{Change} = \frac{0.4}{8} \times 6 = 0.3 \text{ V}$$

8. Ans. 3.10 to 3.12

Sol. Least count = $\frac{0.5}{50}$ mm

$$\text{diameter of ball} = 2.5 \text{ mm} + 20 \times \frac{0.5}{50} \\ = 2.7 \text{ mm}$$

$$\rho = \frac{M}{V} = \frac{M}{\frac{4}{3}\pi R^3}$$

$$\ln \rho = \ln M - \ln \left(\frac{4}{3}\pi R^3 \right)$$

$$\frac{\Delta \rho}{\rho} \times 100 = \frac{\Delta M}{M} \times 100 + 3 \frac{\Delta R}{R} \times 100 \\ = 3.1 \%$$

9. Ans. 2.25

Sol. $V_1 = \frac{\omega}{k} \quad V_2 = \frac{\omega}{3k}$
 $\Rightarrow 3V_2 = V_1$

$$3 \frac{1}{\sqrt{\mu_{r_2} \epsilon_{r_2}}} = \frac{1}{\sqrt{\mu_{r_1} \epsilon_{r_1}}}$$

$$\Rightarrow \frac{\epsilon_{r_2}}{\epsilon_{r_1}} = 9 \frac{\mu_{r_1}}{\mu_{r_2}} = \frac{9}{4} = 2.25$$

10. Ans. 3.73

Sol. $x = \frac{K}{T}$

$$\frac{x_1}{x_2} = \frac{T_2}{T_1} \Rightarrow x_2 = \frac{x_1 T_1}{T_2}$$

$$x_2 = \frac{3.53 \times 10^{-4} \times 373}{353}$$

$$x_2 = 3.73 \times 10^{-4}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(B)

2. Ans.(C)

Sol. In (C) bond order increases during process
 Diatomic systems with ≤ 14 elements
 s-p orbital mixing further

3. Ans.(C)

Sol. In Graph (I) - two radial nodes are present

$$\therefore n - l - 1 = 2 \quad n = 5 \text{ for } l = 2$$

In Graph (II) one radial node is present

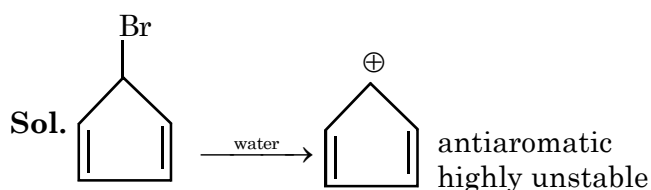
$$\therefore n - l - 1 = 2 \quad n = 4 \text{ for } l = 2$$

In Graph (III) zero radial node

$$\therefore n - l - 1 = 2 \quad n = 3 \text{ for } l = 2$$

Graph (IV) does not represents any radial distribution curve

4. Ans.(D)

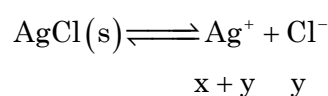
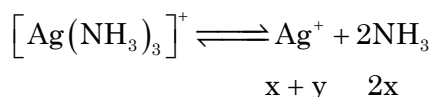


5. Ans.(C)

Sol. Electron affinity of F^+ is more than Cl^+ since
 E.A. of cation is reverse of IE

6. Ans.(B)

Sol. In simultaneous solubility $[Ag^+]$ remains
 same in both equilibrium



$$\text{So } K_c = \frac{(x+y)(2x)^2}{[Ag(NH_3)_3]^+}$$

$$K_{sp} = (x+y)y$$

$$\frac{K_c}{K_{sp}} = \frac{(2x)^2}{[Ag(NH_3)_2]^+ \cdot y}$$

given $(2x) = [NH_3] = 1M$ and

$[Ag(NH_3)_2]^+ = [Cl^-] = y$ because Ag^+ from

$AgCl$ passes in $[Ag(NH_3)_2]^+$

Hence $\frac{k_c}{k_{sp}} = \frac{1}{y \cdot y}$

or $y^2 = \frac{1.8 \times 10^{-10}}{6.2 \times 10^{-8}} = 0.29 \times 10^{-2}$

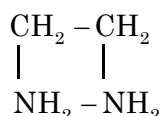
$y = 0.54 \times 10^{-1} = 0.054M$

7. **Ans. (A)**

8. **Ans. (C)**

Sol. $[\text{Co(en)}_2\text{Cl}_2]\text{Cl}$

en = ethylenediamine



Since two chloride must present as ligands hence option (C) is correct

9. **Ans. (B)**

Sol. $\Delta G^0 = -RT \ln K_p$

10. **Ans. (B)**

11. **Ans. (C)**

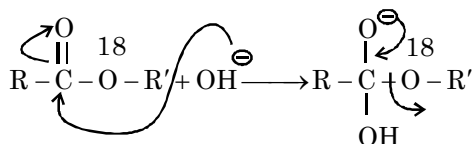
Sol. Cassiterite $\Rightarrow \text{SnO}_2$

12. **Ans. (A)**

Sol. $\Delta^0(\text{NaOH}) = \Delta^0(\text{NaCl}) + \Delta^0(\text{H}_2\text{O}) - \Delta^0(\text{HCl})$
 $= 200 + 140 - 180 = 160 \text{ Scm}^2\text{mol}^{-1}$

13. **Ans. (D)**

Sol.



14. **Ans. (B)**

Sol. Group - 3 cations $\rightarrow \text{Cr}^{+3}, \text{Fe}^{+3}, \text{Al}^{+3}$

Reagent $\rightarrow \text{NH}_4\text{Cl} / \text{NH}_4\text{OH}$

15. **Ans. (D)**

Sol. In liquid phase

$$P_T = P_B^0 \chi_B + P_T^0 \chi_T = 120 \times 0.6 + 50 \times 0.4$$

$$= 72 + 20 = 92 \text{ mmHg}$$

$$\text{In vapour phase } \chi_T = \frac{P_{\text{Toluence}}}{P_{\text{Total}}} \frac{20}{92} \approx 0.22$$

16. **Ans. (A)**

Sol. Hinsberg's test is used for separation of 1°, 2° and 3° Amines.

17. **Ans. (B)**

Sol. Meq of $\text{KMnO}_4 = \text{Meq of H}_2\text{O}_2$

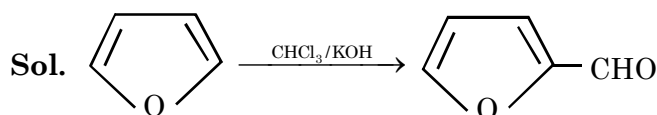
$$16 \times 0.1 \times 5 = 24 \times x \times 2$$

$$x = \frac{16 \times 0.1 \times 5}{24 \times 2} = 0.167M$$

$$\text{Vol strength} = 11.2 \times \text{Molarity} \Rightarrow 11.2 \times 0.167$$

$$= 1.86$$

18. **Ans. (B)**



remaining will give aromatic 6 membered substance

19. **Ans. (A)**

Sol. Gold no expressed in terms of milligram so the amount of starch required to prevent coagulation on addition of 1 ml 10% NaCl to 10 ml gold sol = 0.0025 gm

$$= 2.5 \text{ mg}$$

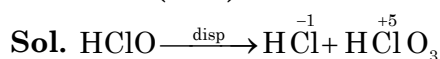
$$\text{So gold No} = 2.5$$

20. **Ans. (B)**

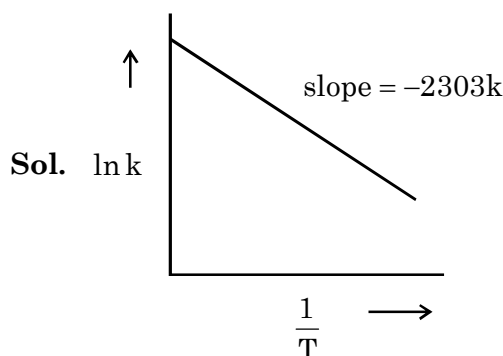
Sol. Bakelite is phenol-formaldehyde resin.

SECTION-II

1. **Ans. (5.00)**



2. **Ans. (1.28)**



Sol. $\ln k$

from Arrhenius equation

$$\ln k = \ln A - \frac{E_a}{R} \times \frac{1}{T}$$

$$\therefore \frac{E_a}{R} = 2303$$

$$\text{or } E_a = 2303R$$

$$\text{Now } \log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\log \frac{k_2}{2 \times 10^{-4}} = \frac{2303R}{2.303R} \left[\frac{1}{400} - \frac{1}{600} \right]$$

$$\log \frac{k_2}{2 \times 10^{-4}} = 10^3 [0.0025 - 0.0017]$$

$$\log \frac{k_2}{2 \times 10^{-4}} = 10^3 \times 0.0008$$

$$\log \frac{k_2}{2 \times 10^{-4}} = 0.8$$

$$\frac{k_2}{2 \times 10^{-4}} = 6.4$$

$$\Rightarrow k_2 = 1.28 \times 10^{-3} \text{ sec}^{-1}$$

3. **Ans.(3.00)**

Sol. Ibuprofen, Aspirin and Phenacetin are antipyretics

4. **Ans.(25.50)**

Sol. For $\text{XeF}_2 \Rightarrow \text{Xe} : \text{F}_2 = 1 : 0.50$

For $\text{XeF}_4 \Rightarrow \text{Xe} : \text{F}_2 = 1 : 5$

For $\text{XeF}_6 \Rightarrow \text{Xe} : \text{F}_2 = 1 : 20$

Sum = 20 + 5 + 0.50 = 25.50

5. **Ans (9.22 to 9.24)**

$$\text{Sol. } P_c V_c = \frac{3}{8} n R T_c$$

$$2 \times V_c = \frac{3}{8} \times 2 \times 0.0821 \times 300$$

$$V = \frac{3}{8} \times 3 \times 8.21 = 9.23$$

6. **Ans.(14.00)**

Sol. Borax = $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$

Kernite = $\text{Na}_2\text{B}_4\text{O}_7 \cdot 4\text{H}_2\text{O}$

7. **Ans.(8.00)**

Sol. In all the ores metal in +2 except in Ag_2S (Argentite) metal is in +1 and in Cassiterite Sn is in +4.

8. **Ans.(9.20)**

9. **Ans.3.00**

$$\Delta S_{\text{system}} = m.s. \ln \frac{T_2}{T_1} = 100 \times 4.2 \times \ln \left(\frac{350}{300} \right) = 63 \text{ J K}^{-1}$$

$$\Delta S_{\text{surrounding}} = -\frac{q_{\text{sys}}}{T_{\text{final}}} = -\frac{m.s.(T_2 - T_1)}{T_2}$$

$$= -\frac{100 \times 4.2 \times (350 - 300)}{350} = -60 \text{ J K}^{-1}$$

$$\therefore \Delta S_{\text{total}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = 63 + (-60) = 3 \text{ J K}^{-1}$$

10. **Ans.(8.00)**

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (C)**

Sol. Any point on the line $x = y = z$ is $(\lambda, \lambda, \lambda)$

This point lies on

$$(\sin A)x + (\sin B)y + (\sin C)z = 2d^2 \text{ \&}$$

$$(\sin 2A)x + (\sin 2B)y + (\sin 2C)z = d^2$$

Eliminating d^2

$$\sin A + \sin B + \sin C = 2(\sin 2A + \sin 2B + \sin 2C)$$

$$4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = 8 \sin A \sin B \sin C$$

$$\therefore \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2} = \frac{1}{16}$$

2. **Ans. (A)**

$$\text{Sol. } 2^{\tan\left(x - \frac{\pi}{4}\right)} - 2(2)^{-2\frac{\sin^2\left(x - \frac{\pi}{4}\right)}{\sin\left(\frac{\pi}{2} - 2x\right)}} + 1 = 0$$

$$2^{\tan\left(x - \frac{\pi}{4}\right)} - 2(2)^{\frac{-2\sin^2\left(x - \frac{\pi}{4}\right)}{2\sin\left(\frac{\pi}{4} - x\right)\cos\left(\frac{\pi}{4} - x\right)}} + 1 = 0$$

$$2^{\tan\left(x - \frac{\pi}{4}\right)} - 2(2)^{\frac{-\sin^2\left(x - \frac{\pi}{4}\right)}{-\sin\left(x - \frac{\pi}{4}\right)\cos\left(\frac{\pi}{4} - x\right)}} + 1 = 0$$

$$2^{\tan\left(x - \frac{\pi}{4}\right)} - 2(2)^{\tan\left(x - \frac{\pi}{4}\right)} + 1 = 0$$

$$2^{\tan\left(x - \frac{\pi}{4}\right)} = 1 = 2^0 \Rightarrow \tan\left(x - \frac{\pi}{4}\right) = 0$$

$$x = \frac{\pi}{4}$$

but $x \neq \frac{\pi}{4}$ no solution.

3. Ans. (A)

Sol. $f(x) = \frac{\pi}{2} + \tan^{-1} x + \tan x + \sin x + \cos x$

$-1 \leq x \leq 1$, $f(x)$ is increasing function

$$m = f(1) = \frac{\pi}{2} + \frac{\pi}{4} + \tan 1 + \sin 1 + \cos 1$$

$$m = f(-1) = \frac{\pi}{2} - \frac{\pi}{4} - \tan 1 - \sin 1 + \cos 1$$

4. Ans. (C)

Sol. As diagonals are bisectors of angle so

$$\text{their equations are } \frac{x-y+k}{\sqrt{2}} = \pm \frac{7x-y+k}{5\sqrt{2}}$$

As they pass through $(1, 2) \Rightarrow k = 0, 5/2$

5. Ans. (B)

Sol. $x^2 + y^2 - 3x - 4y - 32 + \lambda(2x + 6y - 8) = 0$

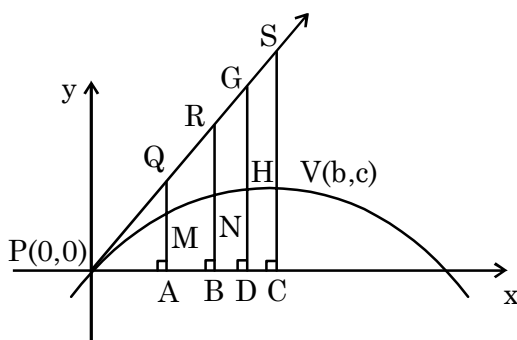
$$S + \lambda L = 0$$

Centre will lie on R.A (property)

6. Ans. (C)

Sol. On transforming, parabola is,

$$(x-b)^2 = -4a(y-c) \text{ and } (b^2 = 4ac)$$



$$PS \equiv y = \frac{b}{2a}x$$

$$\text{For } H(x, y), GH = \frac{x^2}{4a}$$

$$\Rightarrow QM : RN : SK \equiv (2)^2 : (2+5)^2 : (2+5+7)^2$$

7. Ans. (A)

Sol. $\frac{x^2}{a^2+2} + \frac{y^2}{a^2+5} = 1$

$$\frac{2(a^2+2)}{\sqrt{a^2+5}} = 4$$

$$2(a^2+2) = 4\sqrt{a^2+5}$$

$$a^4 + 4 + 4a^2 = 4a^2 + 20$$

$$a^4 = 16$$

$$a^2 = 4$$

$$\frac{x^2}{6} + \frac{y^2}{9} = 1$$

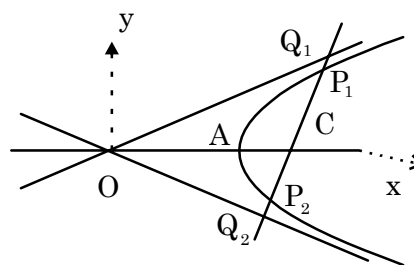
$$e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{6}{9}} = \frac{1}{\sqrt{3}}$$

8. Ans. (C)

Sol. Let $C(x_1, y_1)$ be the mid-point of P_1P_2 , so that the equation of the line P_1P_2 is

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2}$$

It meets the asymptotes $y = \pm \frac{b}{a}x$



$$\text{at } Q_1 \left(x_1 + \frac{ay_1}{b}, \frac{b}{a}x_1 + y_1 \right) \text{ and}$$

$$Q_2 \left(x_1 - \frac{ay_1}{b}, -\frac{b}{a}x_1 + y_1 \right)$$

9. Ans. (B)

Sol. $|z_1 + z_2|^2 + |z_2 + z_3|^2 + |z_3 + z_1|^2$

$$= 2(|z_1|^2 + |z_2|^2 + |z_3|^2) +$$

$$(z_1\bar{z}_2 + \bar{z}_1z_2 + z_2\bar{z}_3 + \bar{z}_2z_3 + z_3\bar{z}_1 + \bar{z}_3z_1)$$

$$= 24 + (z_1\bar{z}_2 + \bar{z}_1z_2 + z_2\bar{z}_3 + \bar{z}_2z_3 + z_3\bar{z}_1 + \bar{z}_3z_1)$$

$$\text{also } |z_1 + z_2 + z_3|^2 \geq 0$$

$$\Rightarrow z_1\bar{z}_2 + z_2\bar{z}_1 + z_2\bar{z}_3 + z_3\bar{z}_2 + z_1\bar{z}_3 + z_3\bar{z}_1 \geq -12$$

$$\therefore |z_1 + z_2|^2 + |z_2 + z_3|^2 + |z_3 + z_1|^2 \geq 12$$

10. Ans. (A)

Sol. Multiply with $\frac{1}{11}$, then shif and subtract twice

$$S = \frac{13 \times 121}{1000} = \frac{1573}{1000}$$

11. Ans. (A)

Sol. $x^3 + ax^2 + bx + c = 0$ — α, β, γ
 $(x - \alpha)(x - \beta)(x - \gamma) = x^3 + ax^2 + bx + c$
 put $x = 1$
 $1 + a + b + c = (1 - \alpha)(1 - \beta)(1 - \gamma) = -385$
 $(1 - \alpha), (1 - \beta), (1 - \gamma)$ are going to negative integers, only possible triplet is $\{-5, -7, -11\}$
 $\alpha = 6 \quad \beta = 8 \quad \gamma = 12$
 $-a = 26$ {sum of roots}

12. Ans. (D)

Sol. Three digits occur twice each
 $\frac{6!}{2!2!2!} \times 10 = 900$
 Two digits occur three each
 $\frac{6!}{3!3!} \times 10 = 200$
 One digit occurs four times and the other twice $2 \times \frac{6!}{4!2!} \times 10 = 300$
 Finally all digits are the same. There are 5 such numbers.
 Thus the total number of admissible numbers is $900 + 200 + 300 + 5 = 1405$

13. Ans. (D)

Sol. As we know $A(\text{adj } A) = |A| I_n$
 $(\text{adj } A) \cdot (\text{adj adj } A) = |\text{adj } A| I_n = |A|^{n-1} I_n$
 $\text{adj adj } A (\text{adj adj adj } A) = |\text{adj adj } A| \cdot I_n = |A|^{(n-1)^2} I_n$
 $\Rightarrow \underbrace{(\text{adj adj adj } \dots \text{adj } A)}_{(n-1)\text{ times}} \cdot \underbrace{(\text{adj adj adj } \dots \text{adj } A)}_{n\text{ times}}$
 $= |A|^{(n-1)^{(n-1)}} I_n$

14. Ans. (A)

Sol. $P\left(\frac{R}{A}\right) = \left(\frac{2}{3}\right)^n \cdot P\left(\frac{R}{B}\right) = \left(\frac{1}{3}\right)^n$, in n throws
 $P(R) = \frac{1}{2} \left(\frac{2}{3}\right)^n + \frac{1}{2} \left(\frac{1}{3}\right)^n$, in n throws
 $P\left(\frac{A}{R}\right) = \frac{P\left(\frac{R}{A}\right) P(A)}{P(R)} = \frac{\frac{1}{2} \left(\frac{2}{3}\right)^n}{\frac{1}{2} \left(\frac{2}{3}\right)^n + \frac{1}{2} \left(\frac{1}{3}\right)^n}$
 $= \frac{2^n}{2^n + 1} = \frac{32}{33} \Rightarrow n = 5$

15. Ans. (D)

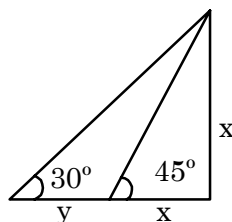
Sol. $n(A) = 2, n(B) = 4$
 $\Rightarrow n(B \times A) = 4 \times 2 = 8$
 $\therefore \text{Relation } (R) \subset B \times A$
 $\Rightarrow \text{Total number of subsets of } B \times A = 2^8 = 2^{2^3}$

16. Ans. (D)

Sol. If $A = I, B = O, A R B$
 $B = O, A = I, B R A, (A, A) \notin R$

17. Ans. (B)

Sol. $\frac{1}{\sqrt{3}} = \frac{x}{x+y}$



$y = (\sqrt{3} - 1)x$ now use unitary method

18. Ans. (A)

Sol. If n is even, the greatest co-efficient is ${}^n C_{n/2}$.
 $\therefore$ the greatest term $= {}^n C_{n/2} x^{n/2}$
 $\Rightarrow {}^n C_{n/2} x^{n/2} > {}^n C_{\frac{n}{2}-1} x^{\frac{n}{2}-1}$ and ${}^n C_{n/2} x^{n/2} > {}^n C_{\frac{n}{2}+1} x^{\frac{n}{2}+1}$
 $\Rightarrow \frac{n - \frac{n}{2} + 1}{\frac{n}{2}} x > 1$ and $\frac{n - \left(\frac{n}{2} + 1\right) + 1}{\frac{n}{2} + 1} x < 1$
 $\Rightarrow x > \frac{\frac{n}{2}}{\frac{n}{2} + 1}$ & $x < \frac{\frac{n}{2} + 1}{\frac{n}{2}} \Rightarrow x > \frac{n}{n+2}$ & $x < \frac{n+2}{n}$
 Hence $\frac{n}{n+2} < x < \frac{n+2}{n}$.

19. Ans. (C)

Sol. $C - (B \cap C)$

20. Ans. (C)

Sol. $\sum_{i=1}^n (x_i + 1)^2 = 9n \Rightarrow \sum x_i^2 + 2 \sum x_i + n = 9n$
 $\sum x_i^2 + 2 \sum x_i = 8n$ (i)
 $\sum_{i=1}^n (x_i - 1)^2 = 5n \Rightarrow \sum x_i^2 - 2 \sum x_i + n = 5n$
 $\sum x_i^2 - 2 \sum x_i = 4n$ (ii)

From (i) and (ii)

$$\sum x_i^2 = 6n \text{ and } \sum x_i = n$$

hence standard deviation

$$= \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2} = \sqrt{\frac{6n}{n} - \left(\frac{n}{n}\right)^2} = \sqrt{5}$$

SECTION-II

1. Ans. 7.00

Sol. At $x = \pm \frac{3}{2}, \pm 1, \pm \frac{1}{2}, 0$,

$f(x) = \text{diff} + \text{non-diff} = \text{non-diff}$

2. Ans. 833.33

Sol. $\int_a^x e^{f(t)} dt = \int_0^x g(x-t) dt + 2x + 3$

$$\Rightarrow \int_a^x e^{f(t)} dt = \int_0^x g(t) dt + 2x + 3$$

differential both sides, we get

$$e^{f(x)} = g(x) + 2$$

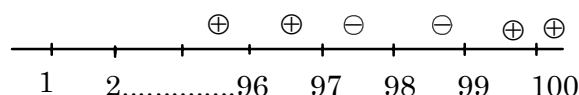
$$\Rightarrow g(x) = e^{f(x)} - 2$$

$$\Rightarrow g'(x) = e^{f(x)} \cdot f'(x)$$

$\therefore e^{f(x)}$ is always greater than zero

$\therefore$ sign of $g'(x)$ is same as sign of $f'(x)$

$\therefore$ sign of $g'(x)$



Clearly, local extremum (maximum or minimum) will occur at $x = 99, 97, 95, 3, 1$

$\therefore$ Sum of all the values $= 1 + 3 + 5 + \dots +$

$$99 = \frac{50}{2} [2 \times 1 + (50-1) \times 2] = 2500$$

3. Ans. 10.00

Sol. $I = \int \left(1 + \frac{3}{x-4} - \frac{24}{x-5} + \frac{30}{x-6} \right) dx$

4. Ans. 5.00

Sol. We have

$$I_n = \int_{-1}^1 |x| \left(1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^{2n}}{2n} \right) dx$$

$$= 2 \left[\frac{x^2}{1.2} + \frac{x^4}{2.4} + \frac{x^6}{4.6} + \dots + \frac{x^{2n+1}}{2n(2n+2)} \right]_0^1$$

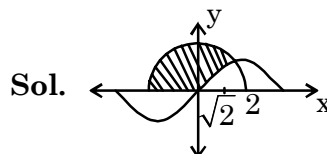
$$= 2 \left[\frac{1}{1.2} + \frac{1}{2.4} + \frac{1}{4.6} + \dots + \frac{1}{2n(2n+2)} \right]$$

$$= 1 + \frac{1}{2} \left[\left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1} \right) \right]$$

$$\text{Hence } \lim_{n \rightarrow \infty} \left[1 + \frac{1}{2} \left(1 - \frac{1}{n+1} \right) \right] = \frac{3}{2}$$

$$\therefore p = 3, q = 2$$

5. Ans. 3.00



Sol.

Area to the left of y -axis $= \pi$

Area to the right of y -axis

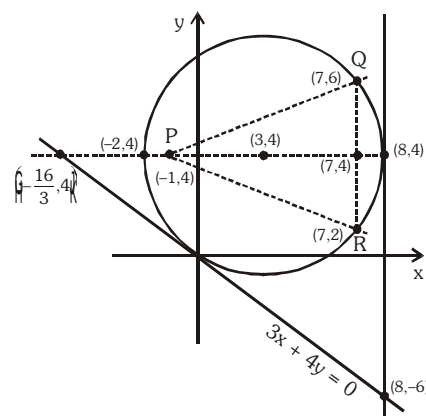
$$= \int_0^{\sqrt{2}} \left(\sqrt{4-x^2} - \sqrt{2} \sin \left(\frac{\pi x}{2\sqrt{2}} \right) \right) dx$$

$$= \left(\frac{x\sqrt{4-x^2}}{2} + \frac{4}{2} \sin^{-1} \frac{x}{2} \right)_0^{\sqrt{2}} + \left(\frac{4}{\pi} \cos \frac{\pi x}{2\sqrt{2}} \right)_0^{\sqrt{2}}$$

$$= 1 + \pi/2 - 4/\pi$$

6. Ans. 200.00

Sol.



$$x^2 + y^2 - 6x - 8y < 0$$

$$(x-3)^2 + (y-4)^2 - 25 < 0$$

Point at least distance from

$(-2, 4)$ is $P(a, b) \equiv P(-1, 4)$

Points which are greatest distance from

$(-2, 4)$ are $Q(c, d)$

& $R(e, f) \equiv Q(7, 6)$ & $R(7, 2)$

ΔPQR is an isosceles triangle & internal bisector of $\angle P$ is $y = 4$

Equation of tangent at origin is $3x + 4y = 0$

equation of tangent at $\left(\frac{c+e}{2} + 1, b \right) \equiv (8, 4)$ is $x = 8$

Area of the right angled triangle formed

$$\text{by above three lines is } \Delta = \frac{1}{2} \times 10 \times \frac{40}{3} = \frac{200}{3}$$

$$\therefore 3\Delta = 200$$

7. Ans. 6.00

$$\begin{aligned} \text{Sol. } \lim_{x \rightarrow 0} \frac{x^3}{\sin^{-1}(x \cos x - \sin x \sqrt{1-x^2})} \\ = \lim_{x \rightarrow 0} \frac{x^3}{x \cos x - \sin x \sqrt{1-x^2}} \quad \left(\because \lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} = 1 \right) \\ = \lim_{x \rightarrow 0} \frac{x^3}{x \left\{ 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right\} - \left\{ x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right\} \sqrt{1-x^2}} \\ = \lim_{x \rightarrow 0} \frac{x^3}{x \left\{ 1 - \sqrt{1-x^2} \right\} - \frac{x^3}{2!} + \frac{x^3}{3!} + \text{higher power of } x} = 6 \end{aligned}$$

Alternate Solution :

$$\begin{aligned} y &= \sin^{-1}(x\sqrt{1-\sin^2 x} - \sin x \sqrt{1-x^2}) \\ \text{put } \sin x &= t \\ \Rightarrow y &= \sin^{-1}(x\sqrt{1-t^2} - t\sqrt{1-x^2}) \\ &= \sin^{-1} x - \sin^{-1} t = \sin^{-1} x - x \\ \Rightarrow \lim_{x \rightarrow 0^+} \frac{x^3}{\sin^{-1} x - x} &= \lim_{x \rightarrow 0^+} \frac{3x^2}{\frac{1}{\sqrt{1-x^2}} - 1} \\ &= \lim_{x \rightarrow 0^+} \frac{3x^2 \sqrt{1-x^2}}{1 - \sqrt{1-x^2}} = \frac{3\sqrt{1-x^2}}{\frac{1 - \sqrt{1-x^2}}{x^2}} = \frac{3}{\frac{1}{2}} = 6 \end{aligned}$$

8. Ans. 34.00

Sol. O (0,0), A(1,0) & B (-1, 0)

Let P (x,y)

$$\overrightarrow{PA} = (1-x)\hat{i} - y\hat{j}$$

$$\overrightarrow{PB} = -(1+x)\hat{i} - y\hat{j}$$

$$\overrightarrow{PA} \cdot \overrightarrow{PB} + 3 \overrightarrow{OA} \cdot \overrightarrow{OB} = 0$$

$$\Rightarrow (x^2 - 1) + y^2 - 3 = 0$$

$$x^2 + y^2 = 4 \quad \dots (1)$$

$$|\overrightarrow{PA}| |\overrightarrow{PB}|$$

$$= \sqrt{(x-1)^2 + y^2} \sqrt{(x+1)^2 + y^2}$$

$$= \sqrt{5-2x} \cdot \sqrt{5+2x}$$

$$= \sqrt{25-4x^2}, \quad x \in (-2, 2) \text{ (from (1))}$$

$$\text{so } M = 5, m = 3 \Rightarrow M^2 + m^2 = 25 + 9 = 34$$

9. Ans. 5.00

Sol. Let A : First 3 cards are king

B : Fourth card is a king

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Let k : drawing of a king

B can occur in four types

Type 1 : $\bar{k}\bar{k}\bar{k}k \rightarrow 1$

Type 2 : $k\bar{k}\bar{k}k \rightarrow 3$

Type 3 : $k\bar{k}k\bar{k} \rightarrow 3$

Type 4 : $kkk\bar{k} \rightarrow 1$

$$\begin{aligned} P(A/B) &= \frac{\frac{1}{{}^{52}C_4}}{\frac{{}^{48}C_3 \times {}^4C_1}{4 \times {}^{52}C_4} + \frac{{}^{48}C_2 \times {}^4C_2 \times 3}{{}^{52}C_4 \times 6} + \frac{{}^{48}C_1 \times {}^4C_3 \times 3}{{}^{52}C_4 \times 4} + \frac{1}{{}^{52}C_4}} \\ &= \frac{1}{20825} \\ \therefore a &= 5 \end{aligned}$$

10. Ans. 9.00

$$\text{Put } \sqrt{x} = \sin \theta, \quad x > 0, \theta \in \left(0, \frac{\pi}{2}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right) + \cot^{-1}\left(\frac{1-\sin\theta}{\cos\theta}\right) = \frac{\pi}{2}$$

$$\Rightarrow \tan^{-1}\left(\tan\frac{\theta}{2}\right) + \cot^{-1}\left(\tan\left(\frac{\pi}{4} - \frac{\theta}{2}\right)\right) = \frac{\pi}{2}$$

$$\Rightarrow \frac{\theta}{2} = \frac{\pi}{2} - \cot^{-1}\left(\tan\left(\frac{\pi}{4} - \frac{\theta}{2}\right)\right) \quad \left(\because \frac{\theta}{2} \in \left(0, \frac{\pi}{4}\right)\right)$$

$$\Rightarrow \frac{\theta}{2} = \tan^{-1}\left(\tan\left(\frac{\pi}{4} - \frac{\theta}{2}\right)\right) \quad \left(\because \frac{\pi}{4} - \frac{\theta}{2} \in \left(0, \frac{\pi}{4}\right)\right)$$

$$\Rightarrow \frac{\theta}{2} = \frac{\pi}{4} - \frac{\theta}{2}$$

$$\Rightarrow \theta = \frac{\pi}{4}$$

$$\Rightarrow \sqrt{x} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow x = \frac{1}{2}$$